



BHABHA UNIVERSITY

DEPARTMENT OF ELECTRONICS AND COMMUNICATION
B.TECH VI SEM

BRANCH: ELECTRONICS AND COMMUNICATION ENGINEERING

S.No.	Subject Code	Subject Name	Periods per week			Credits	Maximum Marks (Theory Slot)			Maximum Marks (Practical Slot)		Total Marks
			L	T	P		End Sem. Exam	Mid sem Tests	Assignments /Quiz	End Sem. Practical /Viva	Practical Record/ assignment/Quiz /Presentation	
1	EC-61	Digital Signal Processing	3	1	2	5	70	20	10	30	20	150
2	EC-62	Antenna & Wave Propagation	3	1	2	5	70	20	10	30	20	150
3	EC-63	Data Communication	3	1	2	5	70	20	10	30	20	150
4	EC-64	Microcontroller & Embedded System	3	1	2	5	70	20	10	30	20	150
5	EC- 65	Seminar			2	2					50	50
6	EC- 66	Minor Project II			2	2					100	100
		Total	12	04	8	24	280	80	40	120	230	750



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SYLLABUS

EC- 61 Digital signal Processing

Unit – I

Discrete-Time Signals and Systems

Discrete-time signals, discrete-time systems, analysis of discrete-time linear time-invariant systems, discrete time systems described by difference equation, solution of difference equation, implementation of discrete-time systems, stability and causality, frequency domain representation of discrete time signals and systems.

Unit - II

The z-Transform

The direct z-transform, properties of the z-transform, rational z-transforms, inversion of the z-transform, analysis of linear time-invariant systems in the z- domain, block diagrams and signal flow graph representation of digital network, matrix representation.

Unit - III

Frequency Analysis of Discrete Time Signals

Discrete fourier series (DFS), properties of the DFS, discrete Fourier transform (DFT), properties of DFT, two dimensional DFT, circular convolution.

Unit - IV

Efficient Computation of the DFT

FFT algorithms, decimation in time algorithm, decimation in frequency algorithm, decomposition for 'N' composite number.

Unit – V

Digital filters Design Techniques

Design of IIR and FIR digital filters, Impulse invariant and bilinear transformation, windowing techniques rectangular and other windows, examples of FIR filters, design using windowing.

References:

1. Oppenheim and Schafer: Digital Signal Processing, PHI Learning.
2. Johnny R. Johnson: Introduction to Digital Signal Processing, PHI Learning.
3. Proakis: Digital Signal Processing, Pearson Education.
4. Rabiner and Gold: Theory and Application of Digital Signal Processing, PHI Learning.
5. Ingle and Proakis: Digital Signal Processing- A MATLAB based Approach, Thompson, Cengage Learning.



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SYLLABUS

List of Experiments:

1. Generation, analysis and plots of discrete-time signals.
2. Implementation of operations on sequences (addition, multiplication, scaling, shifting, folding)
3. Implementation of Linear time-invariant (LTI) systems and testing them for stability and causality.
2. Computation and plot of DTFT of sequences, verification of properties of DTFT.
3. Computation and plots of z-transforms, verification of properties of z-transforms.
4. Computation and plot of DFT of sequences, verification of properties of DFT.
5. Computation and plots of linear/circular convolution of two sequences.
6. Computation of radix-2 FFT- Decimation in time and Decimation in frequency.
7. Implementation of IIR and FIR filter structures (direct, cascade, parallel etc).
8. Implementation of various window design techniques (Rectangular, Bartlett, Hann, Hamming etc).



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SYLLABUS

EC- 62 Antennas and wave Propagation

UNIT I

Radiation

Potential function and the Electromagnetic field, potential functions for Sinusoidal Oscillations, retarded potential, the Alternating current element (or oscillating Electric Dipole), Power radiated by a current element, Application to short antennas, Assumed current distribution, Radiation from a Quarter wave monopole or Half wave dipole, sine and cosine integral, Electromagnetic field close to an antenna, Solution of the potential equations, Far-field Approximation.

UNIT II

Antenna Fundamentals

Introduction, network theorems, directional properties of dipole antennas, travelling –wave antennas and effect of feed on standing-wave antennas, two –element array, horizontal patterns in broad-cast arrays, linear arrays, multiplication of patterns ,effect of earth on vertical patterns, Binomial array, antenna gain, effective area.

UNIT III

Types of antennas

log periodic antenna, loop antenna, helical antenna, biconical antenna, folded dipole antenna, Yagi-Uda antenna, lens antenna, turnstile antenna. Long wire antenna: resonant and travelling wave antennas for different wave lengths, V-antenna, rhombic antenna, beverage antenna,

Unit IV

Aperture and slot Radiation from rectangular apertures, Uniform and Tapered aperture, Horn antenna , Reflector antenna , Aperture blockage , Feeding structures , Slot antennas ,Microstrip antennas – Radiation mechanism – Application ,Numerical tool for antenna analysis

Unit V

Propagation of radio waves

Fundamentals of electromagnetic waves, effects of the environment, modes of propagation. Ground wave propagation- Introduction, plane earth reflection, space wave and surface wave, transition between surface and space wave, tilt of wave front due to ground losses.



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Space wave propagation- Introduction, field strength relation, effects of imperfect earth, curvature of earth and interference zone, shadowing effect of hills and buildings, absorption by atmospheric phenomena, variation of field strength with height, super refraction, scattering, tropospheric propagation, fading, path loss calculations. Sky wave propagation- Introduction, structural details of the ionosphere, wave propagation mechanism, refraction and reflection of sky waves by ionosphere, ray path, critical frequency, MUF, LUF, OF, virtual height, skip distance, relation between MUF and skip distance.

References:

1. Jordan and Balmain: Electromagnetic Waves and Radiating System, PHI Learning.
2. Krauss: Antennas and wave propagation, TMH.
3. Balanis: Antenna Theory Analysis and Design, Wiley India Pvt. Ltd.
4. Harish and Sachidananda: Antennas and wave propagation, Oxford University Press.
5. Raju: Antennas and Wave Propagation, Pearson Education.
6. Kennedy: Electronic Communication Systems, TMH.

List of Experiments:

1. To Plot the Radiation Pattern of an Omni Directional Antenna.
2. To Plot the Radiation Pattern of a Directional Antenna.
3. To Plot the Radiation Pattern of a Parabolic Reflector Antenna.
4. To Plot the Radiation Pattern of a Log Periodic Antenna.
5. To Plot the Radiation Pattern of a Patch Antenna.
6. To Plot the Radiation Pattern of a Dipole/ Folded Dipole Antenna.
7. To Plot the Radiation Pattern of a Yagi (3-EL/4EL) Antenna.
8. To Plot the Radiation Pattern of a Monopole/ WHIP/ Collinear Antenna.
9. To Plot the Radiation Pattern of a Broad site Antenna.
10. To Plot the Radiation Pattern of a Square Loop Antenn



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SYLLABUS

EC- 63 DATA COMMUNICATION

Unit-I

Data Communication: Introduction, Components, data representation Serial & Parallel transmission , Modes of data transmission, Line Encoding: Unipolar, Polar, Bipolar, Networks – Protocols and standards – Standards organizations – Line configurations – Topology – Transmission mode – Categories of networks – Inter networks.

Unit-II

OSI model: Functions of the layers. Transmission media: Guided media – Unguided media – Transmission impairment – Performance. Switching Circuit switching , packet switching (virtual circuit and datagram approach), message switching

Unit-III

ERROR CONTROL AND DATA LINK PROTOCOLS

Error detection and correction: Types of errors – Detection – Vertical Redundancy Check (VRC) – Longitudinal Redundancy Check (LRC) – Cyclic Redundancy Check (CRC) – Check sum – Error Correction. Data Link Layer Protocols: Framing , HDLC, ARQ: Stop and Wait, Sliding Window. Efficiency

Unit-IV

NETWORKS

LAN: Project 802 – Ethernet – Token bus – Token ring – FDDI.

MAN: IEEE 802.6 (DQDB) – SMDS.

X.25, FRAME RELAY, ATM AND SONET/, SDH

Unit-V.

NETWORKING DEVICES AND TCP / IP PROTOCOL SUITE

Networking and internetworking devices: Repeaters – Bridges – Gateways – Other devices – Routing algorithms – Distance vector routing – Link state routing.

TCP / IP protocol suite: Overview of TCP/IP.

REFERENCE BOOKS

1. Data and Computer Communication – W. Stallings, Pearson
2. LANs – Keiser, Tata Mc-Graw Hill
3. Data Communication & Networking – B.A. Forouzan, Tata Mc-Graw Hill
4. Internetworking with TCP/IP – VOL-I – D.E. Comer, PHI
5. ISDN and Broad band ISDN with Frame Relay & ATM – W. Stallings, Pearson



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Textbooks:

1. Computer Networks by Tanenbum/PHI.
2. Data Networks: Bertsekas & Gallager.
3. Shay, William A. / “Understanding Data communications & Networks” / VikasPublishing HousePvt. Ltd.

PRACTICAL; RC 63 DATA COMMUNICATION LAB

- 1 TO STUDY VARIOUS MULTIPLEXING TECHNIQUES
- 2 TO STUDY OF NETWORK INTERFACE CARD (NIC)
- 3 TO STUDY OF PARALLEL AND SERIAL TRANSMISSION
- 4 TO STUDY OF NRZ AND RZ CODES
- 5 TO STUDY OF DIFFERENT TYPES OF MODEM.
- 6 TO STUDY OF INTEGRATED SERVICES DIGITAL NETWORK.
- 7 TO STUDY OF TWISTED PAIR, COAXIAL CABLE AND FIBRE OPTIC CABLE.
- 8 TO STUDY OF DIGITAL INTERFACE RS-232.
- 9 TO STUDY DIFFERENT TOPLOGIES.
- 10 TO STUDY LAN USING STAR TOPOLOGY



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EC- 64 MICROCONTROLLER & EMBEDDED SYSTEM

UNIT-I

8051 Interfacing, Applications and serial communication 8051 interfacing to ADC and DAC, Stepper motor interfacing, Timer/ counter functions, 8051 based data acquisition system 8051 connections to RS-232, 8051 Serial communication , Serial communication modes, Serial communication programming, Serial port programming in C.

UNIT II:

Microcontroller 8096 Introduction to 16-bit Microcontroller, functional block-diagram, memory status, complete 8096 instruction set, classification of instruction set, addressing modes, programming examples using 8096, hardware features of 8096, parallel ports, control & status Registers, Introduction to 16/32 bit PIC microcontrollers and DSPIC.

UNIT-III

Introduction to Embedded Systems:

Definition of embedded system, embedded systems vs. general computing systems, history of embedded systems, classification, major application areas, purpose of embedded systems, characteristics and quality attributes of embedded systems, common design metrics, and processor technology: general purpose processor, application specific processor, single purpose processor.

UNIT-IV

Embedded System Architecture:

Von Neumann v/s Harvard architecture, instruction set architecture, CISC and RISC instructions set architecture, basic embedded processor, microcontroller architecture, CISC & RISC examples: 8051, ARM, DSP processors.

UNIT-V

Input Output and Peripheral Devices

Timers and counters, watchdog timers, interrupt controllers, PWM, keyboard controller, analog to digital converters, real time clock.



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Reference Books:

1. Muhammad Ali Mazidi and Janice Gillespie Mazidi, The 8051 Microcontroller and Embedded Systems, Pearson education, 200 Kenneth J. Ayala, The 8051 Microcontroller Architecture, III edition, CENGAGE Learning.

2V. Udayashankara and M.S. Mallikarjunaswamy, 8051 Microcontroller: Hardware, Software & Applications, Tata McGraw - Hill, 2009.

3McKinlay, The 8051 Microcontroller and Embedded Systems - using assembly and C, PHI, 2006 / Pearson, 2006.

4Tim Wilmshurst, Designing embedded system with PIC microcontrollers Principles and applications. 2nd ed. 2011 Bsp books pvt It

5Shibu K V, "Introduction to Embedded System", TMH.

6David E Simon, "An Embedded Software Primer", Pearson education Asia, 2001.

7Steven F. Barrett, Daniel J. Pack, "Embedded Systems" Pearson education, First Impression2008.

EC- 64 MICROCONTROLLER & EMBEDDED SYSTEM LAB

1. Programming using arithmetic, logical and bitmanipulation instructions of 8051.
2. Program and verify Timer/Counter in 8051.
3. Communication between 8051 kit and PC.
4. To study development tools/environment for ATMEL/PIC microcontroller program and Architecture.
5. Write an ALP to generate square of 10Khz using Timer 0.
6. Write an ALP to display a string on LCD.
7. Write an ALP to interface seven segment with 8051 and display 0-9 on it.
8. Write an ALP to interface DC Motor with 8051
- 9 Write an ALP to transmit the data using P1 of 8051
10. Write an ALP to interface 4x4 keyboard with 8051.
11. Write an ALP to interface temperature sensor using 8051
12. Write an ALP to interface the lcd 16x2 to P16f877A